

Name: K.P. Nedhe

Reg No: 261100690024

Cryptography assignment - 4

1. Decrypt the following ciphertext using the Hill cipher.

Ciphertext: HLDTRIHJDXWQCMAB

key = $\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$ use (A=0, B=1, ..., Z=25)

Find the plaintext

0	1	2	3	4	5	6	7	8	9	10	11	12	13
A	B	C	D	E	F	G	H	I	J	K	L	M	N
14	15	16	17	18	19	20	21	22	23	24	25		
O	P	Q	R	S	T	U	V	W	X	Y	Z		

$$\Rightarrow \det(k) = (3 \times 7) - (2 \times 5) = 21 - 10 = 11 \pmod{26}$$

$$11 \times 19 \equiv 1 \pmod{26}$$

$$11 \times 19 = 209$$

$$209 = 26(8) + 1$$

$$209 \pmod{26} = 1$$

$$11^{-1} = 19 \pmod{26}$$

$$k^{-1} = \det^{-1} \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \pmod{26}$$

$$k^{-1} = 19 \begin{bmatrix} 7 & 24 \\ 21 & 3 \end{bmatrix} = \begin{bmatrix} 133 & 456 \\ 399 & 57 \end{bmatrix} \pmod{26} = \begin{bmatrix} 3 & 14 \\ 9 & 15 \end{bmatrix}$$

$$HD = [7, 11] = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 11 \end{bmatrix} = \begin{bmatrix} 21 + 154 \\ 63 + 55 \end{bmatrix} = \begin{bmatrix} 175 \\ 118 \end{bmatrix} = \begin{bmatrix} 19 \\ 14 \end{bmatrix} = T$$

$$DT = [3, 19] = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 19 \end{bmatrix} = \begin{bmatrix} 9 + 266 \\ 27 + 95 \end{bmatrix} = \begin{bmatrix} 275 \\ 122 \end{bmatrix} = \begin{bmatrix} 15 \\ 18 \end{bmatrix} = P$$

$$QI = [16, 8] = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 16 \\ 8 \end{bmatrix} = \begin{bmatrix} 48 + 112 \\ 144 + 40 \end{bmatrix} = \begin{bmatrix} 160 \\ 184 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix} = F$$

$$HT = [7, 9] = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 9 \end{bmatrix} = \begin{bmatrix} 21 + 126 \\ 63 + 45 \end{bmatrix} = \begin{bmatrix} 147 \\ 108 \end{bmatrix} = \begin{bmatrix} 17 \\ 4 \end{bmatrix} = \begin{matrix} R \\ E \end{matrix}$$

$$DX = [3, 23] = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 3 \\ 23 \end{bmatrix} = \begin{bmatrix} 9 + 322 \\ 27 + 115 \end{bmatrix} = \begin{bmatrix} 331 \\ 142 \end{bmatrix} = \begin{bmatrix} 19 \\ 12 \end{bmatrix} = \begin{matrix} T \\ M \end{matrix}$$

$$WQ = [22, 16] = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 22 \\ 16 \end{bmatrix} = \begin{bmatrix} 66 + 224 \\ 198 + 80 \end{bmatrix} = \begin{bmatrix} 290 \\ 278 \end{bmatrix} = \begin{bmatrix} 4 \\ 18 \end{bmatrix} = \begin{matrix} E \\ S \end{matrix}$$

$$CM = [2, 12] = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 2 \\ 12 \end{bmatrix} = \begin{bmatrix} 6 + 168 \\ 18 + 60 \end{bmatrix} = \begin{bmatrix} 174 \\ 78 \end{bmatrix} = \begin{bmatrix} 18 \\ 0 \end{bmatrix} = \begin{matrix} S \\ A \end{matrix}$$

$$AG = [0, 6] = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix} \begin{bmatrix} 0 \\ 6 \end{bmatrix} = \begin{bmatrix} 0 + 84 \\ 0 + 30 \end{bmatrix} = \begin{bmatrix} 84 \\ 30 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \end{bmatrix} = \begin{matrix} G \\ E \end{matrix}$$

Plaintext: TOP SECRET MESSAGE.

2. Consider the Hill cipher key matrix

$$\begin{bmatrix} 24 & 19 \\ 5 & 14 \end{bmatrix} \quad \text{Ciphertext: RXADDDY}$$

find the plaintext for the above ciphertext

$$\Rightarrow \text{decryption} = k^{-1} C \pmod{26}$$

$$\det(k) = (24 \times 14) - (19 \times 5) = 336 - 95 = 241 \equiv 7 \pmod{26}$$

$$\text{inverse} = 7^{-1} \equiv 1 \pmod{26}$$

$$7 \times 15 = 105$$

$$105 = 26(4) + 1$$

$$105 \pmod{26} = 1$$

$$\underline{7^{-1} = 15}$$

$$k^{-1} = 15 \begin{bmatrix} 14 & -19 \\ -5 & 24 \end{bmatrix} = 15 \begin{bmatrix} 14 & 7 \\ 21 & 24 \end{bmatrix} = \begin{bmatrix} 210 & 105 \\ 315 & 360 \end{bmatrix} \pmod{26}$$

$$k^{-1} = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix}$$

$$Rx = [17, 23] = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 17 \\ 23 \end{bmatrix} = \begin{bmatrix} 34 + 23 \\ 51 + 506 \end{bmatrix} = \begin{bmatrix} 57 \\ 557 \end{bmatrix} = \begin{bmatrix} 5 \\ 11 \end{bmatrix} = \begin{matrix} F \\ L \end{matrix}$$

$$Ao = [0, 14] = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 0 \\ 14 \end{bmatrix} = \begin{bmatrix} 14 \\ 308 \end{bmatrix} = \begin{bmatrix} 14 \\ 22 \end{bmatrix} = \begin{matrix} 0 \\ 14 \end{matrix}$$

$$D4 = [3, 24] = \begin{bmatrix} 2 & 1 \\ 3 & 22 \end{bmatrix} \begin{bmatrix} 3 \\ 24 \end{bmatrix} = \begin{bmatrix} 6 + 24 \\ 9 + 528 \end{bmatrix} = \begin{bmatrix} 30 \\ 537 \end{bmatrix} \bmod 26 = \begin{bmatrix} 4 \\ 17 \end{bmatrix} = \begin{matrix} E \\ R \end{matrix}$$

3. Confusion vs Diffusion

Analyze the following classical ciphers with respect to confusion and diffusion. Complete the table and justify your classification based on how each cipher transforms the plaintext

Cipher	does it change the character / symbol value?	does it change the character position?	Confusion / Diffusion	Justification
Shift cipher	Yes	No	Confusion	Substituted via mathematical shift $(P+k)$; character order is unaltered
Multiplicative cipher	Yes	No	Confusion	substituted via modular multiplication $(P \times k)$; character order is unaltered
Affine cipher	Yes	No	Confusion	combines multiplicative & shift structure $((a \cdot P + b) \bmod 26)$; position remains unchanged

transposition cipher	no	Yes	diffusion	characters retain original values, but their positions are scrambled across the text.
----------------------	----	-----	-----------	---

Relationship between substitution and confusion

In a substitution cipher, the position remains the same, but the actual character value changes.

∴ the relationship between the plaintext character and ciphertext character is hidden. Hence substitution mainly provides confusion.

Relationship between Transposition and diffusion

In transposition, the character itself doesn't change
 ABCDE → CADBE.

∴ the characters are moved to different positions. This spreads the information of the plaintext throughout the ciphertext. Hence transposition mainly provides diffusion.

4. Transposition

Encryption: encrypt the message "the arrow is pointing north direction" using,

→ simple

with key = 4

→ writing row by row into a column, then read column by column

T	H	E	A
R	R	O	W
I	S	P	O
I	N	T	I
N	G	N	O
R	T	H	D
I	R	E	C
T	I	O	N

Ciphertext: TR I N E I T H R S N G T R I E D P T N H E D A W O I D D C N

→ keyword

with keyword: ATLAS

order then
from left
to right

A	T	L	A	S
①	⑤	③	②	④
T	H	E	A	R
R	O	W	I	S
P	O	I	N	T
I	N	G	N	O
R	T	H	D	I
R	E	C	T	I
O	N	X	X	X

Ciphertext: TRPIRRDAINNDTXEWIGKCRSTDIIXHOONTEN

→ double transposition

key₁ = BREAK

B	R	E	A	K
②	⑤	③	①	④
T	H	E	A	R
R	O	W	I	S
P	O	I	N	T
I	N	G	N	O
R	T	H	D	I
R	E	L	T	I
O	N	X	X	X

Ciphertext: A I N N D T X T R P I R R O E W I G H L X R S T O I X M O O N T E N

key₂ = L O G

L	O	G
②	③	①
A	I	N
N	D	T
X	T	R
P	I	R
R	O	E
W	I	G
H	L	X
R	S	T
O	I	I
X	H	O
O	N	T
E	N	X

plaintext: N T R R E G X T I O T X A N X P R W H R O X O E I D I J O I L S I M N N .

5. Transposition cipher - decryption

Key = 10

Ciphertext: TIDIE MRNME REDGIO SANOW RSLTP FEEDO
 388NV AHTUD BIENP GDOAE PEIXD ELNSX

⇒ Remove space:

TIDIEMRNMEREDGIOSANOWRSLTPFEEDOS388NVAHTUDBIENPGDOAPEIXD
 ELNSX.

Total number of characters = 60, Key = 10

No. of columns = 10, No. of rows = $60/10 = 6$.

Divide the ciphertext into 10 groups each containing 6 letters:

T	I	D	I	E	M
R	N	M	E	R	E
O	G	O	S	A	N
O	W	R	S	L	T
P	E	E	E	D	O
S	S	R	W	A	A
H	T	W	B	B	E
E	N	A	G	O	D
A	E	P	E	E	A
D	E	N	N	S	X

write these as columns:

T	R	O	O	P	S	H	E	A	D
I	N	G	W	E	S	T	N	E	E
D	M	O	R	E	S	U	P	P	L
I	E	S	S	E	N	D	G	E	N
E	R	A	L	D	U	B	O	I	S
M	E	N	T	O	A	I	D	X	X

read row by row,

plaintext: TROOPS HEADING WEST NEED MORE SUPPLIES
SEND GENERAL DUBOIS MEN TO AID.

→ keyword: SECRET

Ciphertext: ILEDT NHPTR &DIIT MAXSA XERN T ANONI
SNFOT X

→ remove spaces: ILEDTNHPTRS&DIITMAXSA XERN T ANONI SNFOTX.

total letters = 36

keyword length = 6

$$\text{rows} = 36 / 6 = 6$$

I	L	E	D	T	N
H	P	T	R	&	O
I	I	T	M	A	X
&	A	X	E	R	N
T	A	N	O	N	I
S	N	F	O	T	X

Col 1 = ILEDTN

Col 2 = HPTRSO

Col 3 = IITMAX

Col 4 = SAXERN

Col 5 = TANONI

Col 6 = SNFOTX

Then,

^S T	^E H	^C I	^R S	^E I	^T S
A	P	I	A	I	N
N	T	E	X	T	F
O	R	D	E	M	O
N	S	T	R	A	T
I	O	N	X	X	X

read row by row,

plaintext: THIS IS A PLAIN TEXT FOR DEMONSTRATION.

→ keyword: PLAIN

Ciphertext: VEV EAX ARA ATX HGD

→ remove spaces: VEV EAX ARA ATX HGD

total character = 15

keyword length = 5 rows = $15/5 = 3$

	P	L	A	I	N
	⑤	③	①	②	④
col 1 =	VEV				
col 2 =	EAX				
col 3 =	ARA				
col 4 =	ATX				
col 5 =	HGD				

P	L	A	I	N
⑤	③	①	②	④
H	A	V	E	A
G	R	E	A	T
D	A	V	X	X

By reading them row by row,

plaintext: HAVE A GREAT DAY

6. Permutation

find the permutation inverse for:

1) H 3 1 5 2

P =

	1	2	3	4	5
H	3	1	5	2	

1 in the position 3

2 in the position 5

3 in the position 1

4 in the position 4

5 in the position 2

∴ inverse permutation = 3 5 2 1 4

i) 7 4 1 3 5 2 6

$\Rightarrow P = \begin{matrix} & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ & 7 & 4 & 1 & 3 & 5 & 2 & 6 \end{matrix}$

1 in the position 3
2 in the position 6
3 in the position 4
4 in the position 2
5 in the position 5
6 in the position 7
7 in the position 1

$\therefore$ Inverse permutation:

3 6 4 2 5 7 1

ii) 4 1 3 2

$P = \begin{matrix} & 1 & 2 & 3 & 4 \\ & 4 & 1 & 3 & 2 \end{matrix}$

1 in the position 2
2 in the position 4
3 in the position 3
4 in the position 1

$\therefore$ Inverse permutation:

2 4 3 1

iii) 3 5 1 4 2

$P = \begin{matrix} & 1 & 2 & 3 & 4 & 5 \\ & 3 & 5 & 1 & 4 & 2 \end{matrix}$

1 in the position 3
2 in the position 5
3 in the position 1
4 in the position 4
5 in the position 2

$\therefore$ Inverse permutation:

3 5 1 4 2